Chapter 3. Elementary Matrix Operations and Systems of Linear Equations

Row operations, rank and Gaussian elimination

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3.1 Elementary matrix operations and elementary matrices

Motivation. Each row or column operation of Gaussian elimination is multiplication by an invertible elementary matrix.

Definition (Elementary operations)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, an **elementary row [column] operation** is (1) interchanging two rows [columns]; (2) multiplying a row [column] by a nonzero scalar; (3) adding a scalar multiple of a row [column] to another row [column]. These are of **type** 1, 2, 3.

Example 1

For $A = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$: interchanging rows $1, 2$ gives $B = \left(\begin{smallmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$; multiplying column 2 by 3 gives C; adding 4×row 3 to row 1 gives $M = \left(\begin{smallmatrix} 17 & 2 & 7 & 12 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$.

Definition (Elementary matrix)

An $n \times n$ **elementary matrix** is obtained from I_n by one elementary operation; its type is that of the operation.

Theorem 3.1

If B is obtained from $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ by an elementary row [column] operation, then $B = EA$ $[B = AE]$, where E is the $m \times m$ $[n \times n]$ elementary matrix obtained from I_m $[I_n]$ by the same operation. Conversely, if E is elementary, EA $[AE]$ is obtained from A by the corresponding row [column] operation.

Proof [Supplement]. By Theorem 2.13 (row form), row i of EA is $\sum_k E_{ik}$(row k of A), and row i of E is e_i^t unless the operation changed it. Type 1 (swap p, q): row p of E is e_q^t, so row p of EA is row q of A, and vice versa. Type 2 (row p times c): row p of EA is c·row p of A. Type 3 (add c·row q to row p): row p of E is $e_p^t + ce_q^t$, giving row $p + c$·row q. Other rows are unchanged. Columns: transpose, $(AE)^t = E^t A^t$ with E^t elementary. ■

Example 2

In Example 1, $B = EA$ with $E = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, and $C = AE'$ with $E' = \mathrm{diag}(1, 3, 1, 1)$.

Theorem 3.2

Elementary matrices are invertible, and the inverse of an elementary matrix is elementary of the same type.

Proof. The operation giving E is undone by an operation $\bar{E}$ of the same type (swap again; multiply by c^{-1}; add $-c$ times the row). By Theorem 3.1, $\bar{E}E = I_n$; then $\mathsf{L}_{\bar{E}}\mathsf{L}_E = \mathsf{I}$ makes L_E one-to-one, hence invertible (Theorem 2.5), and $E^{-1} = \bar{E}$. ■

3.2 The rank of a matrix and matrix inverses

Motivation. Rank-preserving elementary operations reduce every matrix to $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$, giving row rank = column rank and inverses via $(A \mid I)$.

Definition (Rank of a matrix)

$\operatorname{rank}(A) = \operatorname{rank}(\mathsf{L}_A)$. Thus an $n \times n$ matrix is invertible iff its rank is n (Theorem 2.5 and Corollary 2 of Theorem 2.18).

Theorem 3.3

For linear $\mathsf{T} : V \to W$ between finite-dimensional spaces with ordered bases β, γ: $\operatorname{rank}(\mathsf{T}) = \operatorname{rank}([\mathsf{T}]_\beta^\gamma)$.

Proof [Supplement]. With $A = [\mathsf{T}]_\beta^\gamma$, $\mathsf{L}_A \phi_\beta = \phi_\gamma \mathsf{T}$ (Theorem 2.14). As ϕ_β is onto, $\mathrm{R}(\mathsf{L}_A) = \mathrm{R}(\mathsf{L}_A \phi_\beta) = \phi_\gamma(\mathrm{R}(\mathsf{T}))$, and the isomorphism ϕ_γ preserves dimension. ■

Theorem 3.4

If P, Q are invertible $m \times m$ and $n \times n$, then (a) $\operatorname{rank}(AQ) = \operatorname{rank}(A)$,
(b) $\operatorname{rank}(PA) = \operatorname{rank}(A)$,
(c) $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$.

Proof.

(a) L_Q is onto, so $\mathrm{R}(\mathsf{L}_{AQ}) = \mathsf{L}_A \mathsf{L}_Q(\mathbb{F}^n) = \mathsf{L}_A(\mathbb{F}^n) = \mathrm{R}(\mathsf{L}_A)$.

(b) [Supplement] $\mathrm{R}(\mathsf{L}_{PA}) = \mathsf{L}_P(\mathrm{R}(\mathsf{L}_A))$, and the isomorphism L_P preserves dimension.

(c) Combine (a) and (b). ■

Corollary

Elementary row and column operations preserve rank.

Proof. Theorems 3.1, 3.2, 3.4. ■

Theorem 3.5

The rank of a matrix is the maximum number of linearly independent columns, i.e. the dimension of the span of its columns.

Proof. $\mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{\mathsf{L}_A(e_j)\} = \operatorname{span}\{a_1, \ldots, a_n\}$ by Theorems 2.2 and 2.13(b), where $a_j = Ae_j$ is the j-th column. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{smallmatrix}\right)$: a_1, a_2 independent, $a_3 = a_1 + a_2$, so $\operatorname{rank} A = 2$.

Example 2

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{smallmatrix}\right)$, row operations give $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & -2 & 2 \\ 0 & -1 & 1 \end{smallmatrix}\right)$; clearing row 1 by column operations leaves two independent columns, so rank $A = 2$.

Theorem 3.6

If $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has rank r, then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into

$$D = \begin{pmatrix} I_r & O_1 \\ O_2 & O_3 \end{pmatrix} \qquad (O_i \text{ zero matrices}).$$

Proof. If $A = O$, $r = 0$ and $D = A$. Otherwise, induct on m to reach some D; then $r = \operatorname{rank} D = \operatorname{rank} A$ (Corollary) and $r \le m, n$.

Step 1 (a pivot). Move some $A_{ij} \neq 0$ to $(1,1)$ by interchanges and scale it to 1.

Step 2 (clear). Clear row 1 and column 1 by type 3 operations: $\left(\begin{smallmatrix} 1 & 0 \\ 0 & B' \end{smallmatrix}\right)$, B' of size $(m-1)\times(n-1)$.

Step 3 (recurse). If $m = 1$, done. Otherwise, by induction (trivially if $B' = O$), operations on the last $m-1$ rows and $n-1$ columns bring B' to $\left(\begin{smallmatrix} I_{r-1} & O \\ O & O \end{smallmatrix}\right)$, leaving row and column 1 intact. ■

Example 3

$A = \left(\begin{smallmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 4 & 4 & 8 & 0 \\ 8 & 2 & 0 & 10 & 2 \\ 6 & 3 & 2 & 9 & 1 \end{smallmatrix}\right)$: swapping rows $1, 2$, scaling and clearing give $D = \left(\begin{smallmatrix} I_3 & O \\ O & O \end{smallmatrix}\right)$, so rank $A = 3$.

Corollary 1

If rank $A = r$, there are invertible B $(m \times m)$ and C $(n \times n)$ with $D = BAC$.

Proof. By Theorem 3.6, $D = E_p \cdots E_1 A G_1 \cdots G_q$ with elementary E_i, G_j (Theorem 3.1); take $B = E_p \cdots E_1$ and $C = G_1 \cdots G_q$, invertible by Theorem 3.2. ■

Corollary 2

For any $A \in \mathrm{M}_{m\times n}(\mathbb{F})$:
(a) $\operatorname{rank}(A^t) = \operatorname{rank}(A)$;
(b) the rank of A is the maximum number of linearly independent rows, i.e. the dimension of the row space;
(c) the rows and the columns of A span subspaces of the same dimension, equal to rank A.

Proof.

(a) $D^t = C^t A^t B^t$ with B^t, C^t invertible, so rank A^t = rank D^t = r = rank A (Theorem 3.4).

(b) Rows of A = columns of A^t; Theorem 3.5 and (a).

(c) Theorem 3.5 and (b). ■

Corollary 3

Every invertible matrix is a product of elementary matrices.

Proof. $\operatorname{rank} A = n$ gives $I_n = E_p \cdots E_1 A G_1 \cdots G_q$, so $A = E_1^{-1} \cdots E_p^{-1} G_q^{-1} \cdots G_1^{-1}$, elementary by Theorem 3.2. ■

Theorem 3.7

For linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ (finite-dimensional) and matrices with AB defined:
(a) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{U})$;
(b) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{T})$;
(c) $\operatorname{rank}(AB) \le \operatorname{rank}(A)$;
(d) $\operatorname{rank}(AB) \le \operatorname{rank}(B)$.

Proof.

(a) $\mathrm{R}(\mathsf{UT}) = \mathsf{U}(\mathsf{T}(V)) \subseteq \mathsf{U}(W) = \mathrm{R}(\mathsf{U})$.

(c) Apply (a) to $\mathsf{L}_A \mathsf{L}_B = \mathsf{L}_{AB}$.

(d) $\operatorname{rank}(AB) = \operatorname{rank}(B^t A^t) \le \operatorname{rank}(B^t) = \operatorname{rank}(B)$ by (c) and Corollary 2.

(b) With bases α, β, γ, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$: $\operatorname{rank}(\mathsf{UT}) = \operatorname{rank}(AB) \le \operatorname{rank}(B) = \operatorname{rank}(\mathsf{T})$ by Theorems 3.3, 2.11 and (d). ■

Example 4

(a) $\left(\begin{smallmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & -1 & 1 \end{smallmatrix}\right)$: two independent rows, rank 2.
(b) $\left(\begin{smallmatrix} 1 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & 0 & 0 \end{smallmatrix}\right)$: row 3 = row 1 − row 2, rank 2.
(c) $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$: subtracting gives rows $(1,2,1), (0,-3,-3), (0,3,3)$, rank 2.

The inverse of a matrix. The **augmented matrix** $(A \mid B)$ (A $m \times n$, B $m \times p$) lists the columns of A then of B; $M(A \mid B) = (MA \mid MB)$. If A is invertible, $A^{-1} = E_p \cdots E_1$ (Corollary 3), and these row operations turn $(A \mid I_n)$ into $(I_n \mid A^{-1})$. Conversely, $(A \mid I_n) \rightsquigarrow (I_n \mid B)$ gives $E_p \cdots E_1 A = I$, so $B = A^{-1}$. A zero row in the left half means $\operatorname{rank} A < n$: not invertible.

Example 5

For $A = \left(\begin{smallmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{smallmatrix}\right)$, row-reducing $(A \mid I_3)$ gives

$$A^{-1} = \begin{pmatrix} \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{pmatrix}.$$

Example 6

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$, reducing $(A \mid I_3)$ gives a zero row on the left ($\operatorname{rank} A = 2$): not invertible.

Example 7

$\mathsf{T}(f) = f + f' + f''$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ with inverse $\left(\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. So T is invertible (Corollary 1 of Theorem 2.18) and $\mathsf{T}^{-1}(a_0 + a_1 x + a_2 x^2) = (a_0 - a_1) + (a_1 - 2a_2)x + a_2 x^2$.

3.3 Systems of linear equations: theoretical aspects

Motivation. Solutions of $Ax = 0$ form $\mathrm{N}(\mathsf{L}_A)$, those of $Ax = b$ a translate of it, and $Ax = b$ is solvable iff b lies in the column space.

A system of m equations in n unknowns is $Ax = b$, **coefficient matrix** $A \in \mathrm{M}_{m\times n}(\mathbb{F})$; a **solution** is $s \in \mathbb{F}^n$ with $As = b$; the system is **consistent** if one exists.

Example 1

(a) $x_1 + x_2 = 3$, $x_1 - x_2 = 1$ has the unique solution $(2, 1)$.
(b) $x_1 + x_2 + x_3 = 1$ has infinitely many, e.g. $(1, 0, 0)$, $(0, 0, 1)$.
(c) $x_1 + x_2 = 0$, $x_1 + x_2 = 1$ has no solution.

Definition (Homogeneous system)

$Ax = b$ is **homogeneous** if $b = 0$; it always has the solution 0.

Theorem 3.8

The solution set K of a homogeneous system $Ax = 0$ (m equations, n unknowns) is $\mathrm{N}(\mathsf{L}_A)$, a subspace of $\mathbb{F}^n$ of dimension $n - \operatorname{rank}(A)$.

Proof. $K = \{s : As = 0\} = \mathrm{N}(\mathsf{L}_A)$, and $\dim \mathrm{N}(\mathsf{L}_A) = n - \operatorname{rank} \mathsf{L}_A$ by Theorem 2.3. ∎

Corollary

If $m < n$, then $Ax = 0$ has a nonzero solution.

Proof. $\operatorname{rank} A \le m < n$, so $\dim K > 0$. ∎

Example 2

(a) $x_1 + 2x_2 + x_3 = 0$, $x_1 - x_2 - x_3 = 0$: $\operatorname{rank} A = 2$, so $\dim K = 1$, and $K = \operatorname{span}\{(1, -2, 3)\}$.
(b) $x_1 - 2x_2 + x_3 = 0$: $\dim K = 2$, with basis $\{(2, 1, 0), (-1, 0, 1)\}$.

Theorem 3.9

Let K be the solution set of $Ax = b$ and K_H that of $Ax = 0$. For any solution s of $Ax = b$, $K = \{s\} + K_H = \{s + k : k \in K_H\}$.

Proof. If $w \in K$, then $A(w - s) = b - b = 0$, so $w = s + (w - s) \in \{s\} + K_H$. Conversely $A(s + k) = b + 0 = b$. ∎

Example 3

(a) $x_1+2x_2+x_3 = 7$, $x_1-x_2-x_3 = -4$ has solution $s = (1, 1, 4)$, so $K = \{(1, 1, 4)+t(1, -2, 3) : t \in \mathbb{F}\}$.
(b) $x_1 - 2x_2 + x_3 = 4$: $K = \{(4, 0, 0) + t_1(2, 1, 0) + t_2(-1, 0, 1)\}$.

Theorem 3.10

Let $Ax = b$ have n equations and n unknowns. If A is invertible, the unique solution is $A^{-1}b$. Conversely, if the system has exactly one solution, A is invertible.

Proof. If A is invertible, $A(A^{-1}b) = b$, and $As = b \Rightarrow s = A^{-1}b$. Conversely, a unique solution gives $K_H = \{0\} = \mathrm{N}(\mathsf{L}_A)$ (Theorem 3.9), so A is invertible (Theorems 2.5, 2.18). ■

Example 4

$2x_2+4x_3 = 2$, $2x_1+4x_2+2x_3 = 3$, $3x_1+3x_2+x_3 = 1$ has the coefficient matrix of Example 5 of Section 3.2, so

$$A^{-1}\begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}-\frac{7}{8}\\ \frac{5}{4}\\ -\frac{1}{8}\end{pmatrix}.$$

Theorem 3.11

$Ax = b$ is consistent iff $\operatorname{rank}(A) = \operatorname{rank}(A \mid b)$.

Proof. Consistent $\iff$ $b \in \mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{a_1, \ldots, a_n\}$ $\iff$ $\operatorname{span}\{a_j\} = \operatorname{span}\{a_j, b\}$ $\iff$ equal dimensions (Theorem 1.11, as the first lies in the second) $\iff$ $\operatorname{rank} A = \operatorname{rank}(A \mid b)$ (Theorem 3.5). ■

Example 5

For $x_1 + x_2 = 0$, $x_1 + x_2 = 1$: $\operatorname{rank} A = 1 < 2 = \operatorname{rank}(A \mid b)$: inconsistent.

Example 6

Is $(3, 3, 2) \in \mathrm{R}(\mathsf{T})$ for $\mathsf{T}(a_1, a_2, a_3) = (a_1+a_2+a_3,\ a_1-a_2+a_3,\ a_1+a_3)$? With $A = \left(\begin{smallmatrix}1 & 1 & 1\\ 1 & -1 & 1\\ 1 & 0 & 1\end{smallmatrix}\right)$, $\operatorname{rank} A = 2$ and $\operatorname{rank}(A \mid b) = 3$. So no.

3.3.1 An application: Leontief's input–output models

Closed model. Producers (food, clothing, housing) consume fractions $A_{ij} \geq 0$ of commodity j, columns summing to 1. Incomes p must satisfy the **equilibrium condition** $(I - A)p = 0$, $p \geq 0$. ($b \geq c$ [$b > c$] is entrywise; b is **nonnegative** [**positive**] if $b \geq 0$ [$b > 0$].) Even $Ap \leq p$ forces $Ap = p$, else $\sum_i p_i > \sum_i \sum_j A_{ij}p_j = \sum_j p_j$. For

$$A = \begin{pmatrix}0.40 & 0.20 & 0.20\\ 0.10 & 0.70 & 0.20\\ 0.50 & 0.10 & 0.60\end{pmatrix}, \qquad p = \begin{pmatrix}0.25\\ 0.35\\ 0.40\end{pmatrix}$$

solves it: incomes in the ratio $5 : 7 : 8$.

Theorem 3.12

Let A be an $n \times n$ input–output matrix (nonnegative entries, columns summing to 1) of the form $A = \left(\begin{smallmatrix}B & C\\ D & E\end{smallmatrix}\right)$, with D a positive $1 \times (n-1)$ vector and C a positive $(n-1) \times 1$ vector. Then $(I - A)x = 0$ has a one-dimensional solution space generated by a nonnegative vector.

Proof [Supplement]. (The book cites the proof.) Write $x = (y, x_n)$, $y \in \mathbb{R}^{n-1}$.

Step 1 (reduction). Columns of $I - A$ sum to 0, so the last equation is minus the sum of the others: $(I - A)x = 0 \iff (I - B)y = x_n C$.

Step 2 ($I - B$ is invertible). Column sums of B are $1 - D_j < 1$. If $(I - B)y = 0$, $y \neq 0$, then $|y_i| \le \sum_j B_{ij}|y_j|$; summing, $\sum_i |y_i| \le \sum_j (1 - D_j)|y_j| < \sum_j |y_j|$, a contradiction.

Step 3 (the solution space). The solutions are $x_n((I - B)^{-1}C, 1)$, spanned by $v = (y_0, 1)$ with $(I - B)y_0 = C$.

Step 4 ($y_0 \ge 0$). Suppose $N = \{i : (y_0)_i < 0\} \neq \emptyset$. For $i \in N$, $(y_0)_i = \sum_j B_{ij}(y_0)_j + C_i > \sum_{j \in N} B_{ij}(y_0)_j$ (terms $j \notin N$ are ≥ 0, $C_i > 0$). Hence

$$\sum_{i \in N} |(y_0)_i| < \sum_{j \in N} \Big(\sum_{i \in N} B_{ij} \Big) |(y_0)_j| \le \sum_{j \in N} |(y_0)_j|,$$

a contradiction. So $v \ge 0$. ∎

Open model. With demands $d \ge 0$ and $A_{ij} \ge 0$ the amount of commodity i per unit of j (column sums ≤ 1), we need $x \ge 0$ with $(I - A)x = d$: $x = (I - A)^{-1}d$ if $(I - A)^{-1} \ge 0$ exists, e.g. when $A^k \to O$, as then $(I - A)^{-1} = I + A + A^2 + \cdots$ (Section 5.3). With

$$A = \begin{pmatrix} 0.30 & 0.20 & 0.30 \\ 0.10 & 0.40 & 0.10 \\ 0.30 & 0.20 & 0.30 \end{pmatrix}, \qquad (I - A)^{-1} = \begin{pmatrix} 2.0 & 1.0 & 1.0 \\ 0.5 & 2.0 & 0.5 \\ 1.0 & 1.0 & 2.0 \end{pmatrix},$$

$d = (30, 20, 10)$ is met by $x = (90, 60, 70)$.

3.4 Systems of linear equations: computational aspects

Motivation. Gaussian elimination brings $(A \mid b)$ to its unique reduced row echelon form, from which solutions and column dependencies are read off.

Definition (Equivalent systems)

Two systems are **equivalent** if they have the same solution set.

Theorem 3.13

If C is an invertible $m \times m$ matrix, $(CA)x = Cb$ is equivalent to $Ax = b$.

Proof. $As = b \Rightarrow CAs = Cb$, and $CAs = Cb \Rightarrow As = C^{-1}CAs = C^{-1}Cb = b$. ∎

Corollary

If $(A' \mid b')$ comes from $(A \mid b)$ by finitely many elementary row operations, $A'x = b'$ is equivalent to $Ax = b$.

Proof. $(A' \mid b') = C(A \mid b) = (CA \mid Cb)$ with C a product of elementary, hence invertible, matrices. ∎

Definition (Reduced row echelon form)

A matrix is in **reduced row echelon form** if
(a) every nonzero row precedes every zero row;
(b) the first nonzero entry of each row is the only nonzero entry in its column;
(c) the first nonzero entry of each row is 1 and lies to the right of the first nonzero entry of the preceding row.

Gaussian elimination. *Forward pass:* in the leftmost nonzero column, bring a nonzero entry to the top (type 1), scale it to 1 (type 2), clear below it (type 3); repeat on the lower-right submatrix. *Backward pass:* from the last nonzero row up, clear above each leading 1. For example

$$\begin{cases} 3x_1 + 2x_2 + 3x_3 - 2x_4 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_3 - x_4 = 2 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{array}\right),$$

so $x_1 = 1 - x_3$, $x_2 = 2$, $x_4 = 3$, with x_3 free.

Example 1

(a) The final matrix above is in reduced row echelon form.
(b) $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ is not: column 1 violates (b), and row 3's leading 1 violates (c).
(c) $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is in reduced row echelon form; $\left(\begin{smallmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is not (leading entry 2).

Theorem 3.14

Gaussian elimination transforms any matrix into reduced row echelon form.

Proof [Supplement]. After the forward pass, leading entries are 1, each strictly right of the one above, with zeros below and zero rows at the bottom: (a), (c) hold. The backward pass adds multiples of a row, which is zero left of its leading 1, so leading entries and the zeros left of them are unchanged, and (b) holds. ∎

Reading off the solution. If $(A \mid b)$ is in reduced row echelon form with r nonzero rows and no row $(0 \cdots 0 \mid 1)$ (else no solution), solve for the leading variables and set the $n - r$ free variables to parameters $t_1, \ldots, t_{n-r}$. For

$$\begin{cases} 2x_1 + 3x_2 + x_3 + 4x_4 - 9x_5 = 17 \\ x_1 + x_2 + x_3 + x_4 - 3x_5 = 6 \\ x_1 + x_2 + x_3 + 2x_4 - 5x_5 = 8 \\ 2x_1 + 2x_2 + 2x_3 + 3x_4 - 8x_5 = 14 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 3 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

the general solution is

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$$

Theorem 3.15

Let $Ax = b$ have r nonzero equations in n unknowns, $\operatorname{rank} A = \operatorname{rank}(A \mid b)$, and $(A \mid b)$ in reduced row echelon form. Then (a) $\operatorname{rank} A = r$;

(b) if the procedure gives $s = s_0 + t_1u_1 + \cdots + t_{n-r}u_{n-r}$, then $\{u_1, \ldots, u_{n-r}\}$ is a basis of the homogeneous solution set K_H and s_0 is a solution.

Proof.

(a) Consistency excludes rows $(0 \cdots 0 \mid 1)$, so each of the r rows has a leading 1 in A, in a column e_i. These r columns are independent and the column space lies in $\mathbb{F}^r$, so $\operatorname{rank} A = r$.

(b) All $t_i = 0$ gives $s_0 \in K$. By Theorem 3.9, $K_H = \{-s_0\} + K = \operatorname{span}\{u_1, \ldots, u_{n-r}\}$, and $\dim K_H = n - r$ (Theorem 3.8), so the u_i form a basis. ■

What the reduced row echelon form says. If A has columns a_j and reduced form $B = CA$ (C invertible), then $b_j = Ca_j$ and, L_C being an isomorphism, the b_j satisfy exactly the linear relations of the a_j.

Theorem 3.16

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank $r > 0$ and reduced row echelon form B. Then
(a) B has exactly r nonzero rows;
(b) for each $i \le r$ there is a column $b_{j_i} = e_i$ ($j_1 < \cdots < j_r$);
(c) the columns $a_{j_1}, \ldots, a_{j_r}$ of A are linearly independent;
(d) if column k of B is $d_1e_1 + \cdots + d_re_r$, then $a_k = d_1a_{j_1} + \cdots + d_ra_{j_r}$.

Proof.

(a) $\operatorname{rank} B = \operatorname{rank} A = r$ (Corollary to Theorem 3.4); the nonzero rows of B are independent (each has a 1 where the others have 0), so there are r (Corollary 2 of Theorem 3.6).

(b) Let j_i be the column of the leading 1 of row i; by (b), (c) of the definition, $b_{j_i} = e_i$ and $j_1 < \cdots < j_r$.

(c) $b_{j_1}, \ldots, b_{j_r}$ are independent, and $\mathsf{L}_{C^{-1}}$ preserves independence.

(d) Every column of B lies in $\operatorname{span}\{e_1, \ldots, e_r\}$ (rows $> r$ are zero), so $b_k = \sum d_ie_i = \sum d_ib_{j_i}$; apply C^{-1}. ■

Corollary

The reduced row echelon form of a matrix is unique.

Proof. By Theorem 3.16, j_i is the first column of A not in the span of the preceding columns, and every other b_k has coordinates d_i fixed by the unique relation $a_k = \sum d_ia_{j_i}$. So A determines B. ■

Example 2

$A = \begin{pmatrix} 2&4&6&2&4\\ 1&2&3&1&1\\ 2&4&8&0&0\\ 3&6&7&5&9 \end{pmatrix}$ has reduced form $B = \begin{pmatrix} 1&2&0&4&0\\ 0&0&1&-1&0\\ 0&0&0&0&1\\ 0&0&0&0&0 \end{pmatrix}$. So $\operatorname{rank} A = 3$, columns $1, 3, 5$ of A are independent, $a_2 = 2a_1$ and $a_4 = 4a_1 - a_3$.

Example 3

$S = \{2 + x + 2x^2 + 3x^3,\ 4 + 2x + 4x^2 + 6x^3,\ 6 + 3x + 8x^2 + 7x^3,\ 2 + x + 5x^3,\ 4 + x + 9x^3\}$ has the columns of A in Example 2 as coordinate vectors, so the first, third and fifth polynomials form a basis of $\operatorname{span}(S)$.

Example 4

$V = \{x \in \mathbb{R}^5 : x_1 + 7x_2 + 5x_3 - 4x_4 + 2x_5 = 0\}$ has dimension 4, with basis $\{(-7,1,0,0,0),(-5,0,1,0,0),(4,0,0,1,0),(-2,0,0,0,1)\}$. To extend the independent $S = \{(-2,0,0,-1,-1),(1,1,-2,-1,-1),(-5,1,0,1,1)\} \subseteq V$, row-reduce the matrix with columns S then this basis: pivots $1,2,3,6$, so $S \cup \{(4,0,0,1,0)\}$ is a basis of V.